

Blood Flow Speed Estimation with Optical Coherence Tomography Angiography Images

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Abstract

Estimating blood flow speed is essential in many medical and physiological applications, yet it is extremely challenging due to complex vascular structure and flow dynamics, particularly for cerebral cortex regions. Existing techniques, such as Optical Doppler Tomography (ODT), generally require complex hardware control and signal processing, and still suffer from inherent system-level artifacts. To address these challenges, we propose a new learning-based approach named **OCTA-Flow**, which directly estimates vascular blood flow speed from Optical Coherence Tomography Angiography (OCTA) images that are commonly used for vascular structure analysis. OCTA-Flow employs several novel components to achieve this goal. First, using an encoder-decoder architecture, OCTA-Flow leverages ODT data as pseudo labels during training, thus bypassing the difficulty of collecting ground truth data. Second, to capture the relationship between vessels of varying scales and their flow speed, we design an Adaptive Window Fusion module that employs multiscale window attention. Third, to mitigate ODT artifacts, we incorporate a Conditional Random Field Decoder that promotes smoothness and consistency in the estimated blood flow. Together, these innovations enable OCTA-Flow to effectively produce accurate flow estimation, suppress the artifacts in ODT, and enhance practicality, benefiting from the established techniques of OCTA data acquisition. The code and data are available at <https://github.com/Spritea/OCTA-Flow>.

1. Introduction

Blood flow speed measurement provides insights into vascular health and functions [16, 28, 31, 38] that are critical in medical and physiological research. However, precise blood flow speed measurement is extremely challenging in

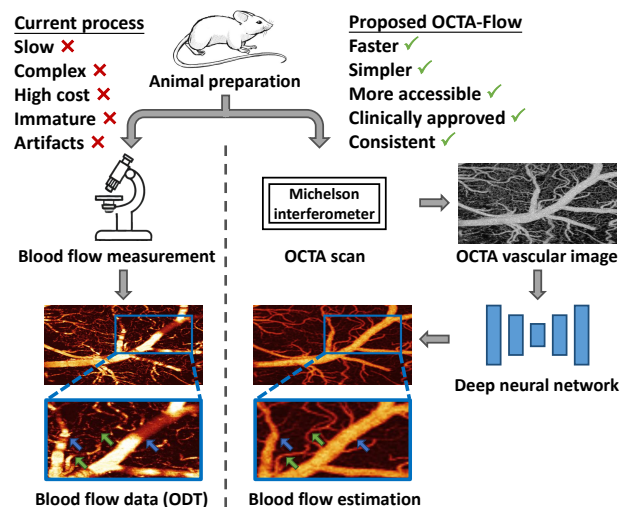


Figure 1. Current blood flow measurement process (left) is slow, complex, costly, immature for clinical use, often suffering from system-level artifacts (highlighted by arrows in zoomed-in panels). In contrast, the proposed OCTA-Flow (right) is faster, simpler, and more accessible by leveraging the clinically approved OCTA technique while mitigating artifacts and generating consistent estimations. Blood flow results are visualized in colormap.

practice [40] due to complex vascular structure and flow dynamics, especially for cerebral cortex regions, where vessel architecture is highly intricate, and sizes vary largely [41].

Existing blood flow (referring to *speed* in this work) measurement techniques are complex and prone to system-level artifacts that affect accuracy [11]. For example, Optical Doppler Tomography (ODT) is costly, requires precise hardware control, and involves complex signal processing [43]. Moreover, ODT is affected by angle artifacts when the light is nearly perpendicular to the vessel, causing artifacts that show drastic flow changes and even disruptions in vessel structures [39, 45] (see blue arrows and green arrows respectively in the zoomed-in panels of Fig. 1).

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Driven by the above observation, we propose a novel solution, named **OCTA-Flow**, which leverages the power of modern deep learning techniques and the practicality of the Optical Coherence Tomography Angiography (OCTA) technique. As shown in Fig. 1, OCTA-Flow directly estimates flow speed from an input OCTA image, suppresses the artifacts in ODT, and enjoys the practicality of established OCTA acquisition techniques.

It is worth noting that, despite being popularly used in clinical and biomedical research for analyzing vascular structures [12, 18], OCTA is not designed for blood flow estimation. At first glance, the vascular structure information in OCTA images provides little/limited clue about the flow speed. However, as we will show later, the prior knowledge of hemodynamics [2, 29] encoded (implicitly and noisily) in OCTA, when combined with carefully designed learning architecture, can serve as an effective flow estimator.

With the above motivation, we develop several novel components in OCTA-Flow for estimating blood flow speed. **First**, OCTA-Flow employs an encoder-decoder architecture for flow estimation. A primary challenge is the absence of ground truth flow data for model training. To address this, we use the noisy flow estimations from ODT as pseudo labels for supervision. Specifically, we construct datasets containing paired high-resolution OCTA and ODT data from the mouse cerebral cortex, which features complex vascular structures. **Second**, an Adaptive Window Fusion (AWF) module is designed to capture the correlation between vessels of varying scales and their flow speed. AWF uses multiple window attention blocks [23] with varying window sizes to extract features at different scales, and then applies automatically generated weights to adaptive feature fusion. This way, it effectively adapts to the vascular distribution across different OCTA images and handles the large variation of different vascular structures. **Third**, a Conditional Random Field Decoder (CRFD) is incorporated to ensure the smoothness in blood flow and meanwhile mitigate artifacts in ODT data (during training). CRFD models the relationship among multilevel features and optimizes both the relationship and features to improve the smoothness of blood flow estimation.

For evaluation, we collected two real datasets *in vivo* from the mouse cerebral cortex. Experiments on both datasets demonstrate that OCTA-Flow not only produces accurate flow estimations but also avoids the angle artifacts present in ODT measurements. We believe this work can inspire researchers to explore this emerging field with significant practical value, as it has the potential to make blood flow measurement more accessible and further improve measurement accuracy.

In summary, our contributions are as follows.

- We propose OCTA-Flow, a novel solution for blood flow estimation that directly predicts blood flow from OCTA

images. In addition to providing high-quality flow estimation, OCTA-Flow bypasses the need for costly and practically challenging measurement techniques like ODT.

- We propose using ODT data as pseudo labels for blood flow of OCTA images during training, addressing the lack of ideal ground truth blood flow speed measurement data.
- We propose an Adaptive Window Fusion module, which captures the correlation between vessels at different scales and their corresponding blood flow speed by multiple window attention blocks and adaptive integration.
- We introduce the Conditional Random Field Decoder to enforce flow estimation smoothness and mitigate ODT artifacts by modeling multilevel feature relationships.
- We build datasets of paired OCTA and ODT images with varied characteristics, which are collected under different animal conditions.
- Experiments show that our method generates accurate flow estimations from OCTA images, outperforms alternative methods, and mitigates artifacts in ODT data.

2. Related Work

2.1. Blood Flow Speed Measurement

Blood flow speed measurement is crucial in biomedical research but typically requires costly hardware, complex operations, and advanced signal processing [11, 40]. Common techniques include Doppler Ultrasound [30], Phase-Contrast MRI [8], and Optical Doppler Tomography [5, 19]. Doppler Ultrasound and MRI detect blood flow in thick tissues but lack the resolution for high-precision measurements. ODT offers high-resolution capillary imaging but suffers from artifacts, leading to inconsistent and underestimated results, limiting clinical use [43, 44].

To bypass these issues of existing blood flow measurements, our work seeks an alternative solution by directly predicting blood flow from OCTA images, while using ODT data as pseudo label during model training.

2.2. Optical Coherence Tomography Angiography

Optical Coherence Tomography Angiography (OCTA) is widely used in clinical and biomedical research [12, 18]. OCTA uses speckle decorrelation to distinguish between vessels and surrounding tissues, where vasculature generally has higher intensity variance across frames than the static tissue, due to the moving red blood cells [33].

Despite its success in vascular structure analysis, OCTA is not designed for blood flow estimation. Typically, a specialized blood flow measurement technique like ODT is additionally used to obtain flow information, where data from both modalities are utilized for physiology study [26]. Although recent works [1, 10] have attempted to estimate blood flow using OCTA, they rely on raw signals with statistical models, require specialized OCT systems, and gener-

ally produce less accurate results than dedicated blood flow measurement techniques. To our best knowledge, our work is the first to directly estimate blood flow speed from OCTA images using generic OCT systems, which is expected to open a new door for the task.

2.3. Deep Regression and Pseudo Labels

Regression is a fundamental task for machine learning. Deep learning-based regression methods have demonstrated their advantage on various regression problems, including age estimation [25], crowd counting [3], human pose estimation [35], and depth estimation [24]. Motivated by the strong learning ability of deep models on complex patterns, we formulate the blood flow estimation in a regression form and develop a deep model to tackle the problem.

Pseudo labels are typically used in problems where the perfect ground truth data is not available, such as weakly-supervised learning problems [20]. Pseudo labels enable models to learn from more samples without ideal labels, improving their performance and generalization. Here we adopt the ODT flow data as the pseudo labels, which contain valuable fine-grained flow information. Despite the presence of artifacts, this approach enables us to leverage the inherent common characteristics of blood flow data.

3. OCTA-Flow

3.1. Problem Formulation

Formally, during inference, given an OCTA image $I \in \mathbb{R}^{H \times W}$ of size $H \times W$, the task of blood flow estimation is to learn a flow estimation model $\mathcal{F}_\theta$ that directly estimates the pixel-wise blood flow speed $Y \in \mathbb{R}^{H \times W}$ by $Y = \mathcal{F}_\theta(I)$, where θ represents parameters of $\mathcal{F}$.

To train the model $\mathcal{F}_\theta$, given a set of K paired OCTA-ODT images $\{I_k, \bar{Y}_k\}_{k=1}^K$, with the ODT data $\bar{Y}_k \in \mathbb{R}^{H \times W}$ as the pseudo label for I_k , the model learns to minimize the discrepancy between $\bar{Y}_k$ and estimated blood flow $Y_k = \mathcal{F}_\theta(I_k)$ by:

$$\min_{\theta} \sum_{k=1}^K \mathcal{L}(\mathcal{F}_\theta(I_k), \bar{Y}_k), \quad (1)$$

where $\mathcal{L}(\cdot)$ is the loss function measuring the discrepancy explained in Sec. 3.5.

3.2. Overview of OCTA-Flow

The overview of the proposed OCTA-Flow is shown in Fig. 2. It has an encoder-decoder like pipeline. A multi-stage backbone network $\mathcal{E}$ (Swin Transformer [23]) serves as the encoder to extract multilevel features, denoted by:

$$\mathbb{B} = \{B_0, B_1, B_2, B_3\} = \mathcal{E}(I), \quad (2)$$

where $B_i \in \mathbb{R}^{H_i \times W_i \times C_i}$, $i \in \{0, 1, 2, 3\}$, are the multilevel features extracted from blocks of different depths. Then the

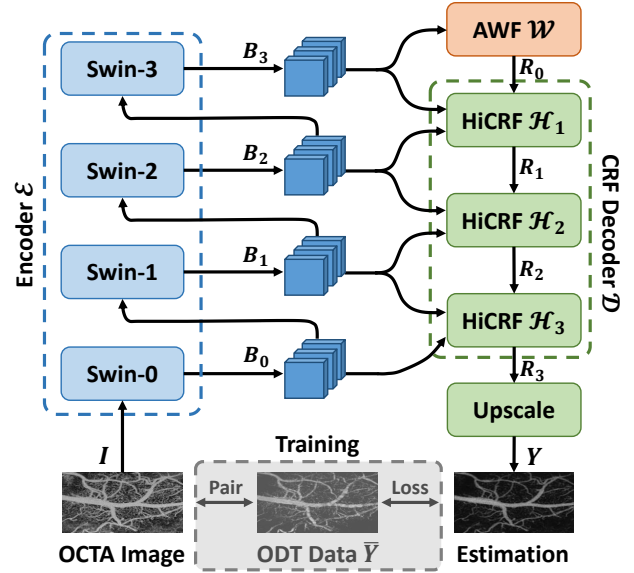


Figure 2. Overview of OCTA-Flow. Our model directly estimates pixel-wise blood flow speed from a given OCTA image during inference. ODT data is only used as the pseudo label during the training process.

deep feature B_3 is fed to the proposed Adaptive Window Fusion module $\mathcal{W}$ to extract adaptive multiscale context information to handle vasculature of various sizes, denoted by:

$$R_0 = \mathcal{W}(B_3) \in \mathbb{R}^{H_3 \times W_3 \times C_0^R}, \quad (3)$$

where C_0^R is the channel number of R_0 . Next, the introduced Conditional Random Field Decoder $\mathcal{D}$ captures the relation of these multilevel features, and enforces smoothness in the prediction results, leading to:

$$R_3 = \mathcal{D}(\mathbb{B}, R_0) \in \mathbb{R}^{H_0 \times W_0}. \quad (4)$$

Finally, R_3 is upsampled with bilinear upsampling to get the pixel-wise blood flow speed estimation $Y \in \mathbb{R}^{H \times W}$ corresponding to the OCTA input image I .

3.3. Adaptive Window Fusion

The vascular system in animals exhibits a remarkable complexity in structure, especially on vessel size. For instance, the mouse cerebral cortex contains vascular structures of varying sizes, such as large pial vessels, medium-sized arteries and veins, and small capillaries. Vessels of different sizes exhibit distinct blood flow speed patterns, since vessel size significantly impacts blood flow speed [2].

To capture the relation between vessels of varying sizes and their blood flow speed, it is essential to incorporate multiscale contextual information. Hence, we propose the Adaptive Window Fusion module, which extracts multiscale context information and integrates them conditioned

on the input adaptively. This adaptive fusion process tailors the feature integration to the specific characteristics of each input, enhancing the ability to capture the relation between diverse vascular structures and the blood flow speed. Fig. 3 shows this module's structure. This module is composed of two parts, one for multiscale context information extraction, and the other for adaptive feature integration.

Window-based Context Extraction. We propose to utilize the window attention mechanism [23] with varying window sizes to capture multiscale context information from the deep features. The small window focuses on local details and captures fine-grained structures, while the large window attends to a wider area and provides more holistic information. Formally, the process is expressed as

$$M_i = \mathcal{W}_i(B_3), i \in \{1, 2, 3\}. \quad (5)$$

Here $\mathcal{W}_i$ is the window attention blocks with window size of $w_i \times w_i$, $i \in \{1, 2, 3\}$. $B_3 \in \mathbb{R}^{H_3 \times W_3 \times C_3}$ is the input deep feature, and $M_1, M_2, M_3 \in \mathbb{R}^{H_3 \times W_3 \times C_0^R}$ are the extracted multiscale context information.

For each window attention block, since the regular window-based attention focuses on each non-overlapping window itself, to bridge and provide connection of windows, the shifted-window attention [23] is further applied following the regular window-based attention. Formally, the window attention block $\mathcal{W}_i$, $i \in \{1, 2, 3\}$, consists of consecutive operations below:

$$\begin{aligned} \widehat{M}_i &= \text{W-MSA}_i(\text{LN}(B_3)) + B_3, \\ \overline{M}_i &= \text{MLP}(\text{LN}(\widehat{M}_i)) + \widehat{M}_i, \\ \widetilde{M}_i &= \text{SW-MSA}_i(\text{LN}(\overline{M}_i)) + \overline{M}_i, \\ M_i &= \text{MLP}(\text{LN}(\widetilde{M}_i)) + \widetilde{M}_i. \end{aligned} \quad (6)$$

Here W-MSA_i, SW-MSA_i refer to the regular window multi-head self-attention and the shifted window multi-head self-attention with window size of $w_i \times w_i$ respectively. LN and MLP refer to layer norm and multilayer perceptron.

This window-based multiscale context information extraction method enjoys multiple key advantages, compared with the widely used pyramid pooling method [48]. (1) Our method keeps more complete information than pyramid pooling. The window attention mechanism maintains information for every element in the input, while the pooling method only keeps local maximum, resulting in inevitable information loss, especially on details like small vessels. (2) As an attention mechanism [37], window attention automatically adjusts the feature weights based on the input vasculature feature, focusing on important regions and features, while the pyramid pooling, as a non-parametric method, uses fixed-scale pooling operations, which are much less adaptive to content and cannot adjust focus accordingly.

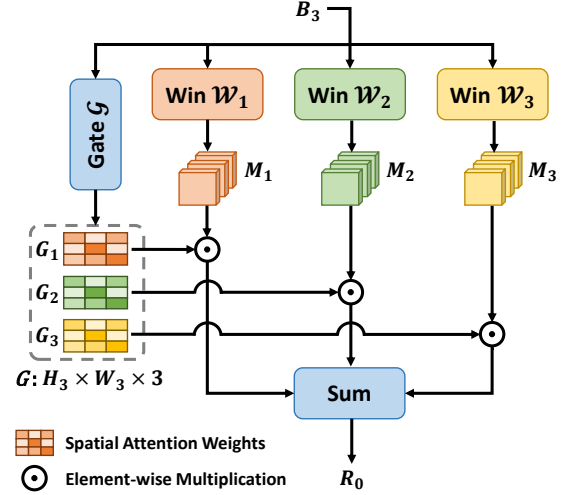


Figure 3. The Adaptive Window Fusion module. This module extracts multiscale context information with window attention, and adaptively integrates them with dynamically generated fine-grained spatial attention weights conditioned on the input.

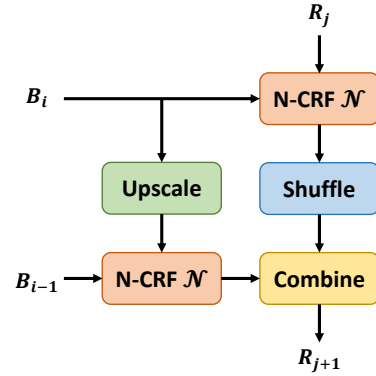


Figure 4. The Hierarchical CRF block. This block models the interdependencies between multilevel features hierarchically, enforcing consistent and smooth blood flow estimation results.

(3) Window attention with shifting enables efficient information flow and interaction across different windows, and blends them naturally. In contrast, pyramid pooling typically extracts features of each pooling window independently, lacking interaction across pooling windows.

Adaptive Feature Integration. The biological and anatomical study [41] shows that the vascular structure differs across regions and individuals, due to metabolic and functional differences. Therefore, we emphasize that the vessel feature integration should be content-aware, depending on the specific vasculature characteristics of the input. Here we propose an adaptive method to integrate multiscale context information dynamically, conditioned on the input vasculature feature. This enables content-aware fusion, which fusion with fixed weights [48] cannot achieve.

Specifically, for the input content B_3 , a convolutional

gate $\mathcal{G}$ is applied to generate the feature weight combination G dynamically. The weight combination is split by channel to obtain three single-channel weights G_1, G_2, G_3 , for multiscale context features respectively. Then the weighted sum is performed to combine all context features, which forms the module's output R_0 . Formally, the process is:

$$\begin{aligned} G &= \mathcal{G}(B_3), \quad (G_1, G_2, G_3) = \text{SP}(G), \\ R_0 &= G_1 \odot M_1 + G_2 \odot M_2 + G_3 \odot M_3. \end{aligned} \quad (7)$$

Here $B_3 \in \mathbb{R}^{H_3 \times W_3 \times C_3}$, $G \in \mathbb{R}^{H_3 \times W_3 \times 3}$, $G_1, G_2, G_3 \in \mathbb{R}^{H_3 \times W_3}$, and SP refers to the channel splitting operation, which extracts every channel of G as the weights. $\odot$ denotes the element-wise multiplication with broadcasting, where $G_i, i \in \{1, 2, 3\}$ is changed from single channel to C_0^R channels by repeating on the channel-axis before performing element-wise multiplication.

Instead of predicting a scalar weight, which multiplies the context feature elements with a same number, we generate a weight matrix G_i . It can be regarded as the spatial attention weight [4], which assigns a unique weight to each element in the spatial dimensions of the context feature. This achieves a more refined attention mechanism that can selectively emphasize or suppress specific regions of the context feature based on the input vasculature feature in the fusion process.

Based on the above process, we provide a robust mechanism for content-aware, multiscale context feature integration. It offers adaptive and fine-grained feature fusion, tailored to the unique characteristics of the input vasculature. Combined with the window-based multiscale context extraction method, they enhance the model's capacity to recognize diverse vascular structures, and capture the correlation between them and the blood flow speed.

3.4. Conditional Random Field Decoder

Conditional Random Field (CRF) [36] is a probabilistic model for structured prediction problems, where the outputs are interdependent rather than independent. It models the relation between the node and its adjacent nodes in a graph, and infers the result by considering these relations comprehensively, leading to consistent and smooth results.

Given that blood flow at a certain position is closely related to its neighbor areas and should exhibit local continuity [2], we propose the Conditional Random Field Decoder (CRFD), inspired by the robust relational modeling capabilities of CRF. The CRF Decoder captures the interdependencies of multilevel features in the form of CRF. By jointly optimizing both the feature relationships and the feature values through backpropagation, this approach enhances the smoothness and accuracy of blood flow estimation.

Hierarchical CRF block. To model the relation of multilevel features, we propose the Hierarchical CRF (HiCRF)

block. Fig. 4 shows its structure. It takes three features from different levels, and models the relation between them progressively and hierarchically. Specifically, given low level feature $B_{i-1} \in \mathbb{R}^{H_{i-1} \times W_{i-1} \times C_{i-1}}$, and middle level feature $B_i \in \mathbb{R}^{H_i \times W_i \times C_i}$ from the backbone network, and high level feature $R_j \in \mathbb{R}^{H_i \times W_i \times C_j^R}$, $j \in \{0, 1, 2\}$, from the previous HiCRF block (or AWF for R_0), we model the relation between B_{i-1} and B_i , and the one between B_i and R_j with the Neural CRF (N-CRF) unit $\mathcal{N}$ respectively. Then the relations are integrated by concatenation and projection as the block's output. Formally, the process is:

$$\begin{aligned} Z_{i,j} &= \mathcal{N}(B_i, R_j), \\ Z'_{i-1,i} &= \mathcal{N}(B_{i-1}, \text{UP}(B_i)), \\ R_{j+1} &= \text{Proj}(Z'_{i-1,i} \oplus \text{PS}(Z_{i,j})). \end{aligned} \quad (8)$$

Here $R_{j+1} \in \mathbb{R}^{H_{i-1} \times W_{i-1} \times C_{j+1}^R}$ is the output of one HiCRF block. UP represents bilinear upsampling, PS represents pixel shuffle, Proj represents projection, and $\oplus$ denotes concatenation by channel. Pixel shuffle is applied to $Z_{i,j}$, due to its deep semantic context from R_j with a large channel count, enhancing sharpness and reducing computational load. Bilinear upsampling is applied to B_i , because it has a mix of semantic and spatial information with a moderate channel count, preserving structural coherence and aligning it spatially with B_{i-1} for later interaction.

Three consecutive HiCRF blocks, $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$, cascade to form the complete CRF Decoder, represented as:

$$\begin{aligned} R_1 &= \mathcal{H}_1(B_3, B_2, R_0), \\ R_2 &= \mathcal{H}_2(B_2, B_1, R_1), \\ R_3 &= \mathcal{H}_3(B_1, B_0, R_2). \end{aligned} \quad (9)$$

The HiCRF blocks benefit the blood flow estimation problem from multiple aspects. (1) It leverages the dependencies between neighbor regions, smooths out flow predictions, and generates more consistent flow results. (2) The integration of multilevel features enables the model to capture both low-level fine-grained structure details, and high-level context with rich semantics. (3) By passing features through cascaded HiCRF blocks, the model iteratively adjusts the features and refines predictions, improving the ability on handling regions with complex flow dynamics.

Neural CRF unit. With the development of deep learning, multiple works implement CRF with neural networks and train the whole network end-to-end [22, 42, 49]. Here we adopt the Neural CRF (N-CRF) implementation [46], due to its ability to model dense pairwise relations comprehensively. It implements fully-connected CRF with window attention [23]. Specifically, the CRF energy function is composed of the unary potential term, which describes the node itself, and the pairwise potential term, which models the dependencies between nodes. The Neural CRF unit imple-

ments the unary potential term with convolution layers, and the pairwise potential term with window attention. The energy function is optimized with the whole model end-to-end by back propagation. Please refer to [46] for details.

3.5. Loss Function

Since the blood flow speed varies a lot for different vessels, simply calculating the absolute error makes the training process dominated by vessels with large blood flow, and ignores those small ones. Besides, the drastic change and extreme values in the pseudo blood flow label caused by the measurement artifacts need to be handled to alleviate its influence on model training.

To address issues above, we adopt a scale-invariant logarithmic loss [7] for model training. Formally, the loss function is defined as:

$$\mathcal{L} = \alpha \sqrt{\frac{1}{N} \sum_i g_i^2} - \frac{\lambda}{N^2} (\sum_i g_i)^2, \quad (10)$$

where $g_i = \log y_i - \log \bar{y}_i$ is the logarithmic difference between the prediction y_i and the pseudo label $\bar{y}_i$. N is the number of pixels in an image. α, λ are constant factors. The logarithmic scaling compresses the blood flow values to similar ranges, and reduces the impact of absolute magnitude differences. It also alleviates the influence of extreme values in the pseudo label due to measurement artifacts and allows for more reliable learning.

3.6. Dataset

To validate this approach, we build datasets with paired OCTA images and ODT blood flow measurements of the mouse cerebral cortex. We use ODT for high-resolution, high-sensitivity blood flow data [34, 43]. The data collection involves three steps: animal preparation, data acquisition, and data processing. Mice require at least 2 months to mature, followed by cranial window surgery and recovery. An ultrahigh-resolution fiber optic OCT system [44] then captures OCTA or ODT data from the same brain region. Raw data is processed with specialized algorithms to generate data volumes, which are projected to 2D images. Note that the OCTA data is 8-bit, while the ODT data is 16-bit, which increases the blood flow speed estimation difficulty.

Because the animal's state (anesthetized or awake) affects imaging characteristics [9], we collected data in both states, creating two separate datasets, Anesthetized Dataset and Awake Dataset. Given the time-consuming and labor-intensive process of OCTA/ODT animal data collection, prior animal studies typically use fewer than 10 samples [13, 14, 21]. We have collected 106 samples—53 pairs of OCTA and ODT data from over 30 animals, covering various conditions. This includes 66 anesthetized and 40 awake samples. Samples of Anesthetized Dataset and

Awake Dataset are shown in Fig. 5. OCTA images and ODT data of Awake Dataset are affected by bulk motion artifacts caused by awake animal [9], appearing as striped lines, while data of Anesthetized Dataset is less affected by such artifacts.

4. Experiments

4.1. Experimental Setup

Implementation details. We adopt Swin Transformer [23] as the model backbone, following previous regression works [32, 46, 47]. w_1, w_2 and w_3 are set as 3, 5 and 7 in the AWF module. α and λ are set as 10 and 0.85 in the loss function respectively. We train the model with 50 epochs. The initial learning rate is 0.0002, which gradually decreases to 0.00002. We use Adam optimizer [15] with β_1 as 0.9, and β_2 as 0.999. We have compared our method with recent advanced regression models of various architectures [6, 17, 27, 32, 46, 47]. We use the official implementations of these methods. All methods are trained and evaluated on both Anesthetized Dataset and Awake Dataset for comparison. We perform five-fold cross validation for comprehensive evaluation on both datasets.

Evaluation metrics. We adopt relative absolute error (Abs Rel), and root mean squared error (RMSE) as the metrics to evaluate methods on this task. These metrics are widely used in the regression task [24]. The metrics are computed based on the model's prediction data with linear scaling, following previous regression works [17, 32]. Abs Rel can handle data with different ranges, and RMSE is more sensitive to large absolute errors. Please refer to the supplementary file for metric details and additional experiments.

4.2. Dataset Evaluation

Anesthetized Dataset. Evaluation results on the Anesthetized Dataset are shown in Table 1 (left part). Our method clearly outperforms the comparison methods. For the Abs Rel metric, it surpasses the second-best model by 0.009, and for the RMSE metric, it outperforms the second-best model by 0.150. These represent significant improvements for these metrics. The low errors, especially Abs Rel of 0.353, show that our model can effectively estimate the blood flow based on the vascular structure, which validates the feasibility of the proposed approach.

Qualitative results of our method are shown in Fig. 5 (top row). ODT measurement is severely affected by the angle artifacts, leading to inconsistent and incorrect results, which appear like alternating bright and black stripes, shown in the green rectangular region. The artifacts can even break vessels, as shown in the blue rectangular region. In contrast, our method can generate consistent and smooth blood flow estimation, with a high alignment with the OCTA vascular structure, thanks to our framework and module design. Our

Table 1. Five-fold cross validation results on Anesthetized Dataset and Awake Dataset. Mean and standard deviation are reported. $\downarrow$ indicates that better performance corresponds to smaller values. Results in bold are the best. Results underlined are the second-best.

Method	Architecture	Anesthetized Dataset		Awake Dataset	
		Abs Rel $\downarrow$	RMSE $\downarrow$	Abs Rel $\downarrow$	RMSE $\downarrow$
BTS [17]	DenseNet	0.374 $\pm$ 0.033	6.777 $\pm$ 0.532	0.366 $\pm$ 0.026	6.336 $\pm$ 0.776
IEBins [32]	Swin	0.377 $\pm$ 0.057	7.217 $\pm$ 0.730	0.364 $\pm$ 0.056	7.958 $\pm$ 1.197
NeuWin [46]	Swin	0.366 $\pm$ 0.046	<u>6.428$\pm$0.337</u>	0.367 $\pm$ 0.038	6.324 $\pm$ 0.759
Ord Ent [47]	Swin	<u>0.362$\pm$0.045</u>	6.442 $\pm$ 0.447	<u>0.359$\pm$0.036</u>	<u>6.315$\pm$0.853</u>
Diff Depth [6]	Diffusion	0.485 $\pm$ 0.037	7.649 $\pm$ 0.341	0.457 $\pm$ 0.070	7.241 $\pm$ 0.736
ECoDepth [27]	Diffusion	0.445 $\pm$ 0.065	7.958 $\pm$ 1.286	0.766 $\pm$ 0.080	7.513 $\pm$ 0.565
OCTA-Flow (ours)	Swin	0.353$\pm$0.042	6.278$\pm$0.480	0.318$\pm$0.018	6.037$\pm$0.674

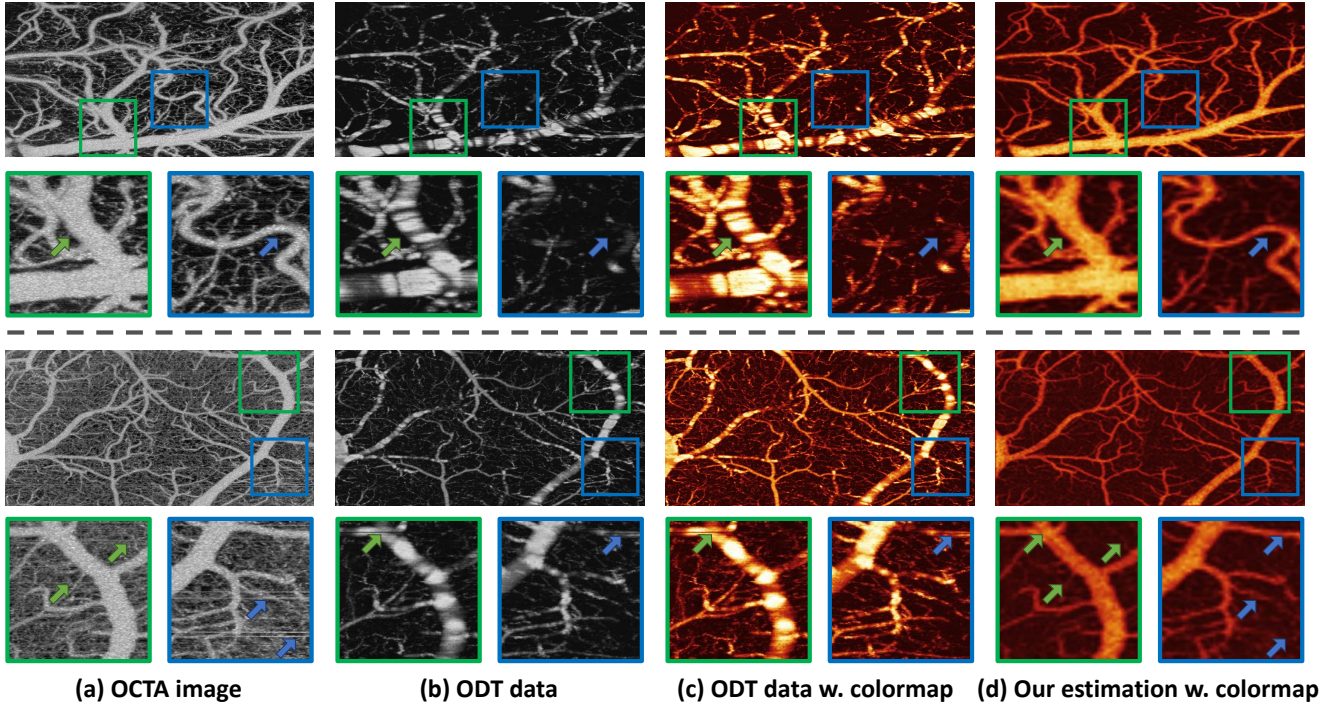


Figure 5. Qualitative results of our method on Anesthetized Dataset (top row) and Awake Dataset (bottom row). Zoomed-in regions with arrows highlight details. Top rows show that our method can generate more continuous and smoother blood flow estimations than ODT data by mitigating the measurement artifacts. Bottom rows show that both OCTA images and ODT data are affected by motion artifacts caused by awake animal’s movements, while our method is robust in handling motion artifacts present in both the OCTA and ODT data.

estimation also shows consistency with hemodynamics [2, 29], where the blood flow speed gradually decreases when the blood flow goes from large vessels to small branches.

Awake Dataset. Evaluation results on the Awake Dataset are shown in Table 1 (right part). Our method outperforms comparison methods as well. The increased advantage (0.041 on Abs Rel and 0.278 on RMSE) of our method over the second-best method highlights its robustness to the motion artifacts on this dataset.

Qualitative results of our method are shown in Fig. 5 (bottom row). The OCTA image is heavily influenced by the motion artifacts in the awake group, leading to white

striped lines across the whole image, as pointed by the arrows in the zoomed-in rectangular regions of column (a). ODT measurement is also affected by the artifacts, appearing as black striped lines, highlighted by the arrows in the zoomed-in rectangular regions of column (b) and (c). Conversely, our method is robust to motion artifacts, providing a clean blood flow estimation. We attribute this to both the AWF module, which captures intricate vascular structures and discriminates vessels from the artifact noise, and the CRF Decoder, which mitigates the impact of outliers by modeling the interdependencies of multilevel features and enforces consistent and smooth blood flow estimations.

Table 2. Ablation study of the main components on Anesthetized Dataset. Base denotes the UNet-like baseline model. AWF is the Adaptive Window Fusion module. CRFD is the CRF Decoder. Δ denotes the performance difference compared with the base model. Results in bold are the best.

Model	Abs Rel $\downarrow$	Δ_{abs}	RMSE $\downarrow$	Δ_{rmse}
Base	0.393	-	7.232	-
Base + AWF	0.344	0.049	6.998	0.234
Base + CRFD	0.358	0.035	7.011	0.221
Base + AWF + CRFD (ours)	0.328	0.065	6.661	0.571

Table 3. Ablation study of the Adaptive Window Fusion module on Anesthetized Dataset. W/O AWF means the model without the AWF module. PPM means replacing the AWF module with the pyramid pooling module. $\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3$ are window attention blocks with different window sizes. $\mathcal{G}$ means using the dynamic weights generated by the gate. Δ denotes the performance difference compared with the model without the AWF module. Results in bold are the best.

Model	Abs Rel $\downarrow$	Δ_{abs}	RMSE $\downarrow$	Δ_{rmse}
W/O AWF	0.358	-	7.011	-
PPM	0.342	0.016	6.964	0.047
$\mathcal{W}_1$	0.348	0.010	6.922	0.089
$\mathcal{W}_1 + \mathcal{W}_2$	0.343	0.015	6.838	0.173
$\mathcal{W}_1 + \mathcal{W}_2 + \mathcal{W}_3$	0.335	0.023	6.790	0.221
$\mathcal{W}_1 + \mathcal{W}_2 + \mathcal{W}_3 + \mathcal{G}$ (ours)	0.328	0.030	6.661	0.350

Table 4. Ablation study of the CRF Decoder on Anesthetized Dataset. W/O CRFD means the model without the CRF Decoder. $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$ refer to the first, second and third HiCRF block. Δ denotes the performance difference compared with the model without CRF Decoder. Results in bold are the best.

Model	Abs Rel $\downarrow$	Δ_{abs}	RMSE $\downarrow$	Δ_{rmse}
W/O CRFD	0.344	-	6.998	-
$\mathcal{H}_1$	0.339	0.005	6.945	0.053
$\mathcal{H}_1 + \mathcal{H}_2$	0.334	0.010	6.884	0.114
$\mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}_3$ (ours)	0.328	0.016	6.661	0.337

4.3. Ablation Study

Main components. Table 2 shows the ablation study results of the main components on Anesthetized Dataset using the default split fold. Base refers to a UNet-like baseline model with skip connections, using the same backbone network as ours. Using AWF or CRF Decoder alone leads to significant performance improvement, and their combination further improves the results. This demonstrates that AWF and CRF Decoder are not only individually effective, but

also have complementary effects for the task.

AWF module. Table 3 presents the ablation study results of the AWF module on Anesthetized Dataset. Using one window attention block improves Abs Rel by 0.010 and RMSE by 0.089, and using multiple window attention blocks with different window sizes further boosts the performance by 0.013 on Abs Rel and 0.132 on RMSE. This validates the effectiveness of our window attention based multiscale context information fusing. Applying dynamic weights conditioned on the input brings totally 0.03 improvement on Abs Rel and 0.35 improvement on RMSE. This shows the benefit of using input-based adaptive fusion.

Compared to the pyramid pooling module, our method performs significantly better than it with the improvement of 0.014 on Abs Rel and 0.303 on RMSE. This demonstrates the advantage of our window attention-based method on capturing complicated vascular structure than the widely used pyramid pooling method.

CRF Decoder. Table 4 summarizes the ablation study of the CRF Decoder on the Anesthetized Dataset. Using a single HiCRF block improves Abs Rel by 0.005 and RMSE by 0.053. When more HiCRF blocks are applied in cascade and more features from multiple levels are included, the performance continues increasing, and achieves an improvement of 0.016 on Abs Rel and 0.337 on RMSE. This study demonstrates the effectiveness of modeling interdependencies across multilevel features in the CRF form, and the necessity of progressive refinement through cascaded HiCRF blocks.

5. Conclusion

We propose a novel approach for estimating blood flow speed directly from OCTA images, circumventing the need of costly blood flow measurements. ODT data is used as pseudo label to address the lack of ideal ground truth measurement data. We design an Adaptive Window Fusion module to capture the correlation between complex vascular structures and the flow speed, and the CRF Decoder to model interdependencies across multilevel features, enforcing smooth and consistent predictions. In addition, we collected two real datasets for evaluation, and experiments on the datasets demonstrate that our method produces accurate flow speed estimation without the artifacts in ODT measurements. We believe this work can inspire future research in this emerging field of significant practical value, with great potential to make blood flow speed measurement more accessible and accurate.

Acknowledgement. This work was supported in part by NIH Grants 2R01 DA029718 (CD, YP), R21DA042597 (CD, YP, HL), NIH’s Intramural Program of NIAAA (NDV) and NSF Grants 2331769 (HL).

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